



POORNIMA

INSTITUTE OF ENGINEERING & TECHNOLOGY

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Basic Mechanical Engineering

251ME3-10/252ME3-10

Notes

Unit-1

Introduction

- ① Mechanical Engineering is an engineering branch that combine engineering physics and mathematics principles with material science to design, analyze, manufacture and maintain mechanical system.
- ② Mechanical Engineering is the study, design, development construction and testing of mechanical and thermal sensors and device, including tools, engines and machines. Mechanical engineer works mostly in engineering service research, development and manufacturing.

Need

without mechanical engineering, not have to things like engine generators, elevators; air conditioning. Mechanical Engineering play a important role in manufacturing technologies from cars to airplanes to refrigerators. It enable to do many activity with ease as it bring helpful technologies to our modern society. It is one of the most important subdivision of Engineering because without it many of the new technologies use every day would not be available.

• Mechanical Engineering called as Mother of engineering

Branches, Because all the present in market, at early stage not developed as much as today. These all are converted stage by stage from beginning to present position.

As mechanical Engineer create solution and solve problems. They are needed on all stages of product from research and development to design and manufacturing through to installation and final commissioning.

Mechanical Engineering touches virtually every aspect of modern life - from mobile phone and biomedical device to aircraft and powerplant. Mechanical Engineer deals with economic issue from cost of single component to economic impact of a manufacturing plant. It also be found sales, management and all engineering aspects.

Basic Mechanical Engineering Subjects :-

- (1) Mathematics
- (2) Statics and Dynamics
- (3) Strength of Materials
- (4) Material Engineering, Composites
- (5) Thermodynamics, heat transfer energy conversion
- (6) Fuels, combustion, I.C. Engine

- (7) Fluid Mechanics
- (8) Mechanism and Machine design (Kinematics & Dynamics)
- (9) Instrumentation and measurement
- (10) Manufacturing Engineering
- (11) Vibration & Control theory
- (12) Hydraulics and Pneumatics
- (13) Mechatronics and Robotics
- (14) Engineering Design and product design
- (15) Computer Aided design (CAD)
- (16) Computer Aided manufacturing (CAM)
- (17) Micro electro mechanical system (MEMS)
- (18) Friction stir welding (FSW)
- (19) Nanotechnology
- (20) Finite Element analysis (FEM)
- (21) Biomechanics
- (22) Computational fluid dynamics
- (23) Acoustical Engineering.

Application

Mechanical Engineering is a diverse subject that derives breadth from need to design and manufacturing everything from small individual parts and device like microscale sensor and inkjet nozzle for printer to the large system. (ie spacecraft and machine tools). Mechanical Engineering deals with anything that moves including the human body a very complex machine. Mechanical Engineer learn about materials, solid and fluid mechanics, thermodynamics, heat transfer, control instrumentation, design and manufacturing to understand mechanical system. Specialized mechanical engineering subjects includes:-

- (1) Biomechanics
- (2) Energy conversion
- (3) Laser assisted material Processing combustion
- (4) Fracture mechanics
- (5) Tribology
- (6) Nuclear Engineering
- (7) Solar Engineering
- (8) Energy Resources

Importance and Scope

- (1) It is one of the broadest and oldest branch.
- (2) It deals with all aspects of conversion of thermal energy into useful work and the machine make it possible
- (3) without mechanical Engineering we would not think like engine, generator, elevator airconditioning and more from car to airplanes
- (4) Mechanical Engineering skill are highly valued in society and their salary reflect that fact.
- (5) Mechanical Engineer are multi skilled as a jack of all trades Mechanical Engineer have a working knowledge of computer application, electricity, structures, mathematics and physics plus they are required some understanding of social environmental and economical factor.
- (6) Global opportunities
- (7) The huge variety of mechanical courses.

Introduction

THERMAL ENGINEERING

Thermal Engineering deals with movements of heat energy and transfer. The energy can be transferred between two medium or transferred into other form of energy. Thermodynamics relate heat, work, temperature & energy. It deals energy transfer from one place to other & one form to another. The heat is a form of energy corresponding to the definite amount of mechanical work.

Thermodynamic state

A thermodynamic state is a system's condition at any given time. These properties are characteristic parameters that have definite value at each state and independent to the way in which system arrived at state. Change in value of property depend only on Initial & final state of system not on the path followed from one state to another. Such property called function.

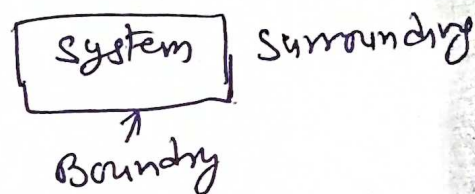
Thermodynamic Equilibrium

In which there is no tendency for the state of a system to change spontaneously, if the system is isolated from surroundings.

Basic Concept of Thermal Engineering

System:- A region in space containing the quantity of matter whose behaviour is being investigated

Everything external to the system is called surrounding. The system is separated by Boundary from surrounding



Closed system:- A system through which only energy (transfer) may cross the system Boundary but mass cannot transfer.

Exp. A quantity of fluid in the cylinder bounded by piston

Open system:- A system through which Both mass and energy may cross the system boundary.

Exp.:- A hot water in open Jar.

Isolated system:- A system through which neither energy nor mass crosses the system Boundary is called Isolated system.

Homogeneous and Heterogeneous System

A system consisting of single phase is called a homogeneous system while a system consisting more than one phase is known as a Heterogeneous system.

Mass:- A quantity of matter inside the body (m)

Weight:- The amount of pull through which the earth pull any body towards its center.

Weight = mass $\times$ gravity

$$W = m \times g. \text{ Newton}$$

force :- Anything that can change the position of object

$$F = ma \text{ By second law of Newton's}$$

Temperature :- The degree of hotness or coldness

$$\frac{C}{5} = \frac{F-32}{9} = \frac{F-273}{5}$$

Pressure :- force per unit area .

$$1 \text{ Bar} = 10^5 \text{ N/m}^2, \text{ In SI pascal unit} = 1 \text{ N/m}^2$$

Energy :- The capacity of doing work .

Kinetic Energy :- Energy of Body due to its motion

$$KE = \frac{1}{2}mv^2$$

Potential Energy :- The energy of Body due to its position above the ground is called potential Energy.

$$PE = mgh$$

Internal Energy :- (U) Energy of a body due to its molecular movement is called Internal energy

$$\text{Total Energy } E = PE + KE + U$$

Intensive property :- are those property of system which does not depend upon the mass as Temperature .

Extensive property :- properties which depend upon the mass of the body are called extensive property as Entropy

Laws of perfect Gas:-

- (1) Boyle's law :- At constant temperature, pressure of given mass of perfect gas is inversely proportional to its volume.

$$P \propto \frac{1}{V}$$

$$PV = \text{constant}$$

- (2) Charles's law :- At constant pressure for a perfect gas volume of given mass is directly proportional to its temperature.

$$V \propto T$$

$$\frac{V}{T} = \text{constant}$$

- (3) Gay Lussac's law :- At constant volume the pressure of given mass of perfect gas is directly proportional to its temperature.

$$P \propto T$$

$$\frac{P}{T} = \text{constant}$$

- (4) Joule's law :- For a perfect gas change in internal energy of a system is directly proportional to the change in temperature.

$$du \propto dt$$

$$du = m c_v dt \quad c_v = \text{specific heat at constant volume.}$$

The ideal gas equation

$$PV = nRT$$

$n = \text{No. of moles.}$

$$\text{or } PV = nMRT$$

$$PV = mRT$$

$R = \text{Universal gas constant } 8.314 \text{ J/mol}\cdot\text{K}$
 $= 259.8 \text{ Nm/kg}\cdot\text{K}$

$$R_0 = 8314.3 \text{ Nm/mol}\cdot\text{K}$$

Specific heat :- specific heat is defined as the amount of heat required to raise the temperature at 1°C of 1 kg substance.

C_p & C_v = Specific heat at constant pressure and volume.

Specific Heat at constant Volume C_v

The rate of change of Internal energy with respect to absolute temperature at constant volume (C_v)

$$C_v = \left(\frac{\partial u}{\partial T} \right)_v \quad u = \text{is internal energy} \\ T = \text{absolute Temp.}$$

$$\begin{aligned} \Phi &= \Delta u + W = \Delta u + P \Delta v \\ &= \int_{T_1}^{T_2} C_v dT \end{aligned}$$

Enthalpy is sum of internal energy and product of pressure and volume

$$h = u + PV$$

Specific Heat at constant Pressure (C_p)

The rate of change of enthalpy with respect to absolute temperature when pressure is constant - (C_p)

$$C_p = \left(\frac{\partial h}{\partial T} \right)_p$$

$$\Delta h = \Delta \Phi = \int_{T_1}^{T_2} C_p dT$$

$$C_p - C_v = \frac{R}{J}$$

$$\text{and } \gamma = \frac{C_p}{C_v}$$

Non flow process

Enthalpy H $\Delta H = \Delta E + P \Delta V$

Entropy S $\Delta S = \frac{\Delta Q}{T}$

Entropy is the amount of intrinsic disorder within the system (No exchange of mass)

Non flow Equation

$$Q = W + \Delta U$$

Non uniform flow: - a flow in which characteristic & parameter vary and different at different location.

(1) Constant Volume process

In this process volume remain constant i.e. $\Delta V = 0$ This also known as isochoric process By first law of thermodynamics

$$Q = \Delta U + W$$

$$W = \int P dV = 0$$

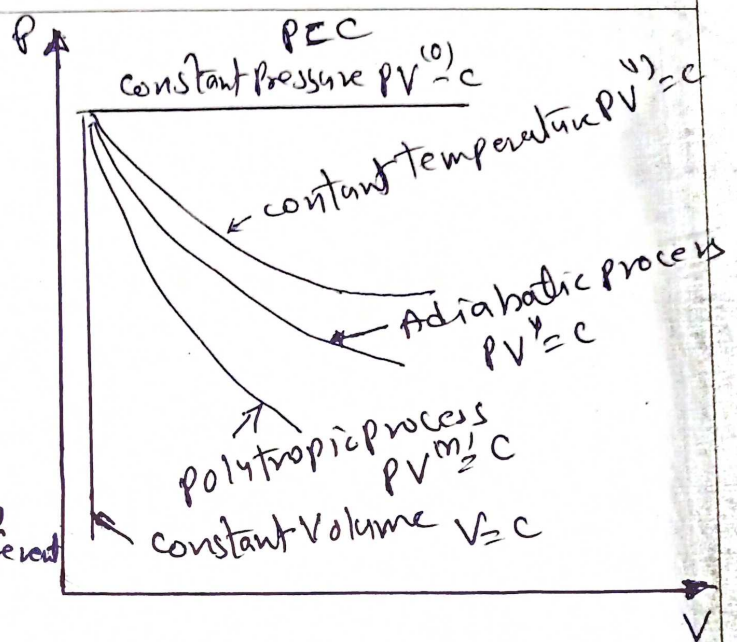
$$Q = \Delta U = m(u_2 - u_1) = m C_V dT \\ = m C_V (T_2 - T_1) \text{ KJ}$$

(2) constant Pressure (Isobaric process)

In this process pressure remain constant - $\Delta P = 0$

$$W = P dV = P (V_2 - V_1)$$

$$\int_1^2 dQ = \int_1^2 dU + \int_1^2 dW$$



$$\begin{aligned}
 Q_{1-2} &= m c_v (T_2 - T_1) + p (V_1 - V_2) \\
 &= m \frac{R}{\gamma - 1} (T_2 - T_1) + m R (T_2 - T_1) \\
 &= \frac{m R (T_2 - T_1) \gamma}{\gamma - 1}
 \end{aligned}$$

(3) constant Temperature process (Isothermal process)

$$pV = p_1 V_1 = p_2 V_2 = \text{constant}.$$

$$\begin{aligned}
 \text{Work } W_{1-2} &= \int_1^2 p dV = \int_1^2 \frac{p_1 V_1}{V} dV = p_1 V_1 \log_e \frac{V_2}{V_1} = p_1 V_1 \log \frac{p_1}{p_2} \\
 &= n R T \log_e \left(\frac{p_1}{p_2} \right)
 \end{aligned}$$

$$Q_{1-2} = W_{1-2} + (U_2 - U_1) = W_{1-2} + 0$$

$$U_2 - U_1 = m c_v (T_2 - T_1) = 0$$

(4) Adiabatic process (

Heat transfer is equal to zero

$$\text{workdone } W_{1-2} = \frac{p_1 V_1 - p_2 V_2}{\gamma - 1}$$

$$\begin{aligned}
 dQ &= dU + dW \\
 Q_{1-2} &= (U_2 - U_1) + \frac{p_1 V_1 - p_2 V_2}{\gamma - 1} \\
 0 &= m c_v (T_2 - T_1) + \frac{p_1 V_1 - p_2 V_2}{\gamma - 1}
 \end{aligned}$$

$$W_{1-2} = m c_v (T_2 - T_1)$$

(5) Polytropic process workdone $W_{1-2} = \frac{p_1 V_1 - p_2 V_2}{n - 1}$
 $pV^n = \text{constant}$

$$Q_{1-2} = m c_v (T_2 - T_1) \frac{(\gamma - n)}{1 - n}$$

Thermodynamic laws

(1) Zeroth law of thermodynamics :-

When two systems are each in thermal equilibrium with third system, the first two systems are in thermal equilibrium with each other.

(2) First law of thermodynamics

It is also known as conservation of energy. The change in a system's internal energy is equal to the difference between heat added to the system from surroundings and work done by the system on the surroundings.

The energy can neither be created nor be destroyed but can be changed from one form to another form.

i.e.

$$Q = \Delta U + W$$

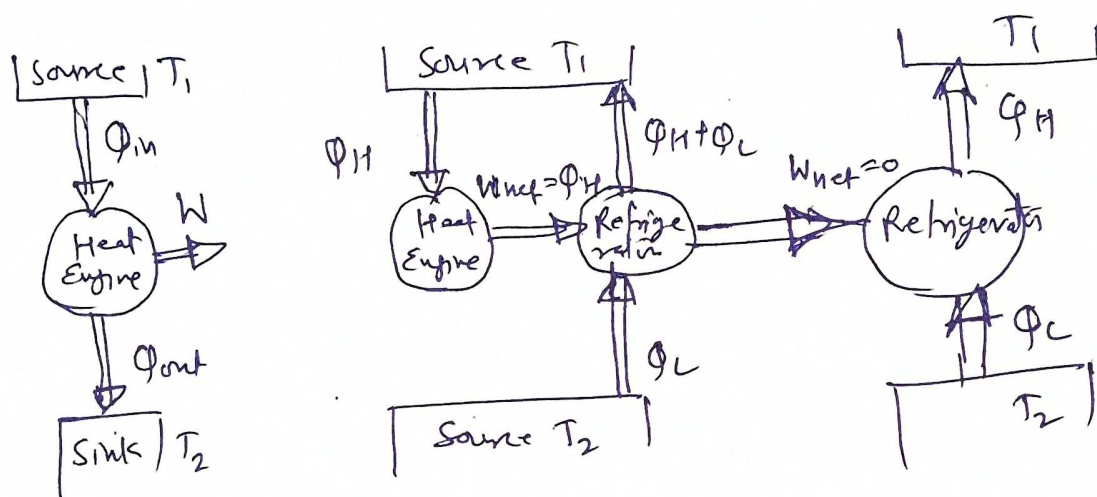
Q : amount of heat added ΔU : Increase in internal energy

W : Work done by the system

(3) Second law of thermodynamics

(a) Kelvin Planck's statement - It is not possible to convert all the heat extracted from a hot body (Q_H) into work (W), without losing some heat (Q_C) to a cold body.

(b) Clausius Statement - It is impossible to transfer heat from a hot body to a cold body without expenditure to work by external energy source.



Third law of thermodynamics

The entropy of perfect crystal of an element in its most stable state form tend to zero as the temperature approaches absolute zero. This allow an absolute scale for entropy to established that from a statistical point of view determine the degree of randomness or disorder in a system.

Applications of thermal engineering

- 1) Boiler design
- 2) Heat exchanger
- 3) Refrigeration system

(4) Solar heating

(5) Thermal Insulation

(6) Thermal power plants.

Heat Engine, Refrigerator and Heat Pump

Coefficient of Performance (COP) is defined as ratio of useful heating or cooling effect of to work required.

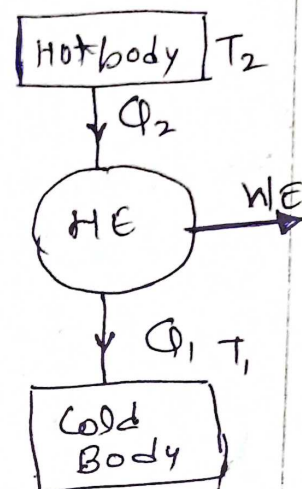
Heat Engine

C.O.P of Heat Engine is express in efficiency.

Heat Engine took Q_2 amount of ~~heat~~ heat from hot body and did work equivalent to W_E

The heat supplied to sink is equal to Q_1

Here useful effect is workdone W_E . It is equal to $Q_2 - Q_1$



C.O.P of heat engine, engine efficiency $\eta = \frac{\text{Workdone}}{\text{Heat supplied}}$

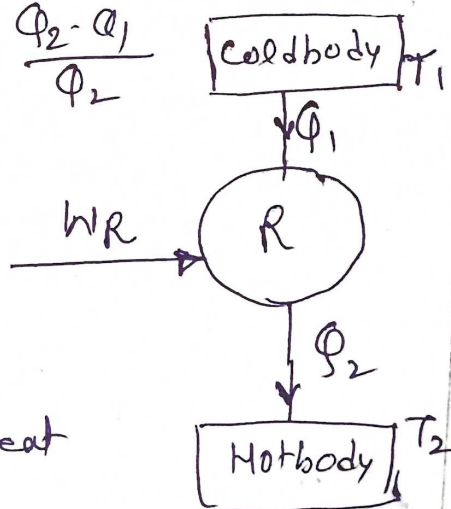
$$\eta = \frac{Q_2 - Q_1}{Q_2}$$

Refrigerator

In Refrigerator Q_1 amount of heat extracted from cold body and deliver ^{Q_2} to amount of heat to the hot body with the help of

Input work W_R . Here $Q_2 = Q_1 + W_R$

In Refrigerator useful effect is extraction of heat Q_1



Then COP of refrigeration is

$$\begin{aligned} \text{COP}_R &= \frac{Q_1}{W_R} \\ &= \frac{Q_1}{Q_2 - Q_1} \end{aligned}$$

Heat pump

Heat pump is similar to refrigerator but in this case the desired effect is heat delivered to Hot body Q_2

The COP of heat pump is

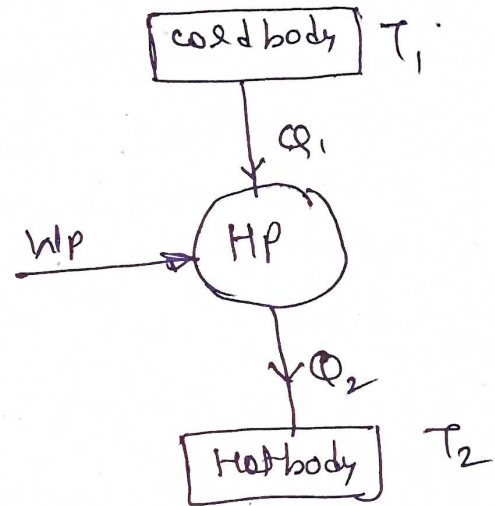
$$\begin{aligned} (\text{COP})_p &= \frac{Q_2}{W_p} \\ &= \frac{Q_2}{Q_2 - Q_1} \end{aligned}$$

The relation between the COP of refrigerator and COP of heat pump.

$$\begin{aligned} (\text{COP})_p &= \frac{Q_2}{Q_2 - Q_1} \\ &= 1 + \frac{Q_1}{Q_2 - Q_1} \end{aligned}$$

$$(\text{COP})_p = 1 + (\text{COP})_R$$

COP of heat pump is always greater than the unity



✓ Machine Design

III

Machine design is an area of study in which one learns to design machine parts by utilizing its whole knowledge and experience in the field of mechanical engineering. Machine design can be defined as creation of right combination of correctly proportioned moving and stationary component so joined and constructed as to enable to liberation, transformation and utilization of energy.

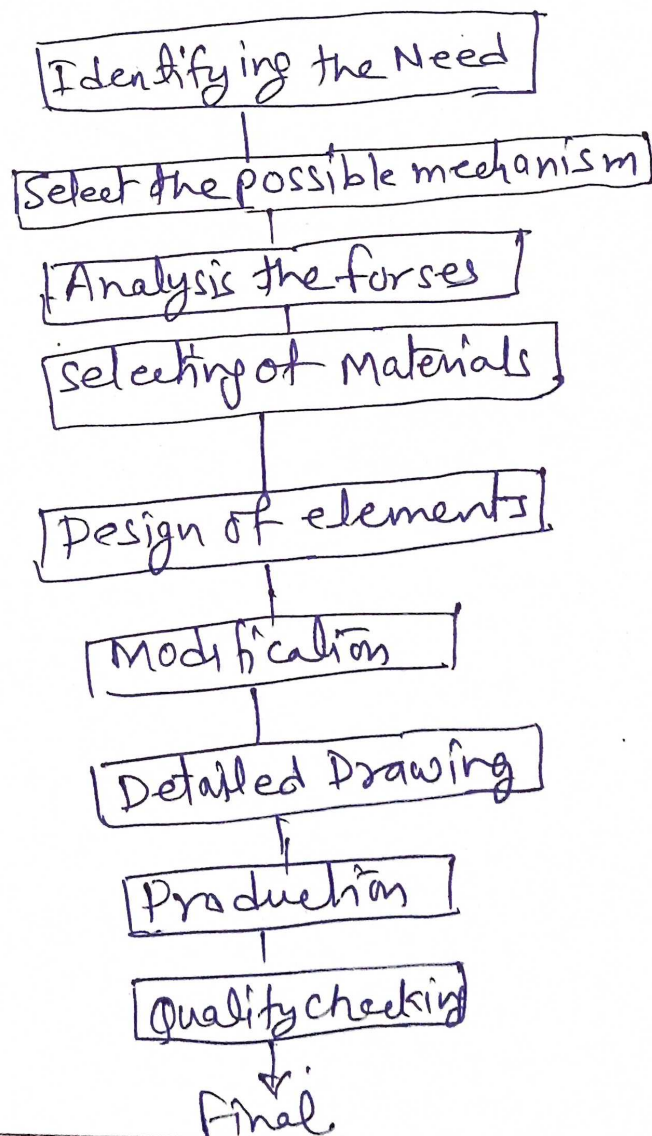
In machine design the designer is to determine the motion, forces & energy transfer involved so to determine the size, shape and material for each element. Design work may involve concentration on one component at a time but simultaneously consider its relation to the other component and product as whole.

In machine design following steps followed:-

- (1) The problem to be solved or desired purpose of machine is completely and clearly stated.
- (2) Possible mechanisms for desired motion selected
- (3) Forces and energy transmitted
- (4) Best suitable material selected
- (5) Allowable value of stress and deflection determine

- (6) Size and shape of each element is determined
- (7) Dimension of machine element are modified.
- (8) Assembly and detailed drawing of machine are made ✓ with complete specification of material and manufacturing method.

Machine design procedure.



✓ Design consideration :- Important factors considered in designing below:-

- (1) Functionality
- (2) Strength
- (3) Stiffness
- (4) Wear
- (5) Corrosion
- (6) Thermal property
- (7) Surface finish
- (8) Lubrication
- (9) Friction
- (10) Weight
- (11) Noise
- (12) Shape
- (13) Size
- (14) Safety
- (15) Reliability
- (16) Manufacturability
- (17) Utility
- (18) Maintenance
- (19) Cost
- (20) Life
- (21) Marketability.
- (22) Control.

✓ Objectives of Design

- (1) Design for strength
- (2) Design for function
- (3) Design for manufacturing
- (4) Design for Safety
- (5) Design for look and Comfort.

✓ constraints of Designing

Design constraints are condition that need to be happen for project to be successful. Design constraint help narrow choice when creating a project.

constraints are limitation of the design. These may be material available, the cost of material, the amount of time they have to develop the solution.

Common constraints are

- (1) cost
- (2) scope
- (3) Quality
- (4) customer satisfaction
- (5) Risk
- (6) resources (7) Time

Design methodology

It is the broader study of method of design the study of principle practice and procedure of designing.

there are five design methodologies:-

- (1) Instructional system design
- (2) Design thinking
- (3) Agile Design
- (4) System thinking
- (5) X-problem

Design methodology in technology.

- (1) Topdown Design
- (2) Bottomup Design
- (3) Structure design
- (4) Data Structure System Development
- (5) Object oriented Design.

Classification of Machine Design

- (1) Adaptive Design:-
Minor alterations in existing design hence less skill required.

(2) Development Design

Product is completely converted into a new product by redesigning & scientific training & design capability required.

(3) New Design

lot of research, technical ability and creativity.

(1) Rational design:- design mathematical formulas.

(2) Impirical :- formulas based on practice and experience

(3) Industrial - production aspects

(4) Optimum design - best design

(5) System design - complex structural like cars.

(6) Element design - element of system like piston

(7) CAD

✓ Classification of material

(1) metals and alloy iron steel copper aluminium etc.

(2) Non metal like glass, rubber, plastic etc.

Selection of Material

(1) Availability

(2) Suitability for working conditions

(3) Cost of material

Physical properties of metals

Density, melting point, Thermal conductivity

Mechanical Properties

- ① Strength :- ability to resist the externally applied forces without breaking
- ② Stiffness :- ability to resist deformation under stress
- ③ Elasticity :- To regain original shape after deformation, when external force removed.
Steel is more elastic than rubber
- ④ Plasticity :- To retain the deformation produced under load permanently.
- ⑤ Ductility :- Draw the wires, strings & plastic wire
- ⑥ Brittleness :- Opposite, Breakdown with little permanent distortion.
- ⑦ Malleability :- It is a property of a material by which it can be beaten to form thin sheets.
Examp :- Zinc, iron, aluminium
- ⑧ Toughness :- To resist fracture due to high impact load like hammer blow.

- ⑨ Machinability:- Property by which a material can be cut measured in number of way
- ⑩ Resilience - Ability to absorb energy and to resist shocks and impact loads.
Amount of energy absorbed/unit volume within elastic limit
- ⑪ Creep:- a part subjected to constant stress at high temperature for a long period of time. Undergoes slow and permanent deformation.
- ⑫ Fatigue:- material subjected to repeated stresses it failed stresses below the yield stress. progressive crack formation.
- ⑬ Hardness:- Resistance to wear, scratching, deformation

Iron $\rightarrow$ cast, wrought, steel and alloy steel.

Cast Iron - 2% or more carbon

Steel - low carbon < 0.3

medium 0.3 to 0.8

High 0.8 to 1.5

it is also called tool steel or cast steel.

Types of Manufacturing process

- ① Primary shaping process - casting, forging, extrusion, drawing, bending etc.
- ② Machining process -  Turning, planing, shaping
- ③ Surface finishing process - polishing, honing, lapping, electroplating, superfinishing.
- ④ Joining process - welding, riveting, soldering.

Industrial Engineering

IV

It is concerned with design, analysis and controls of production and service operation and system

Industrial Engineering is the application of engineering principle and techniques of scientific management to maintenance of a high level of productivity at optimum cost in Industry.

So the Industrial Engineering is the branch of engineering that involved figuring out how to make or do things better. Industrial Engineer are concerned with reducing production cost increasing efficiency improving quality of product and service, ensuring workers health and safety, protecting the environment and complying the government regulations.

They work to eliminate waste of time money materials energy and other commodities.

Common term used in Industrial Engineering:-

- (1) Computer & Information system
- (2) Cost Engineering & project Management
- (3) Distribution & Marketing

- (4) Distribution & Marketing
- (5) Employee & Industrial Relation
- (6) Economy
- (7) Ergonomics
- (8) Management
- (9) Manufacturing system
- (10) Material Processing
- (11) Health & Safety
- (12) Operation & Inventory planning & control
- (13) Quality Assurance & Reliability
- (14) Organization planning
- (15) Work design and Management

Plant Layout

Plant layout in production planning/management is the physical arrangement of building, machinery, equipment, work place and other facilities of production in order to process the product in the most effective manner.

Best layout are integration, minimum movements and material handling. Smooth and continuous flow, safe and improved environment and flexibility.

Following are three types of layout

- (1) Process (2) Product (3) Fixed position layout

(1) Process layout - It is also known as functional layout or analytical layout. This type of layout is employed where:

(1) low volume of production

(2) similar job manufacture

(3) machines are arranged in on function basis.

It has following drawbacks:

(1) More floor space

(2) production control difficult and costly.

(3) Routing & Scheduling is difficult

(4) Handling & Back tracking

(2) Product layout - It is also known as synthetic layout. This type of layout is best suitable where one type of product is produced and the product is standardized. It is used for mass production of product. The product layout lowers overall manufacturing cost, time, required less space for placing machine and utilizes machine and labour best. In this specialized and strict supervision required and machine cannot be used to their maximum capacity. This product layout is suitable for automobile manufacturing concern.

(3) Fixed position layout - It is also known as static product layout. In this product is so large that it cannot be moved from one location to another location but tools, material and labour are moved to the product to perform

specific operation. This type of layout ~~is~~ is used for manufacture of ship, aeroplanes, steam turbine etc.

Flow system or flow charts

(1) Operation process chart - (OPC) It represents the sequence of operations to be performed on a component.

(2) Flow charts - It is generally used in process industry to indicate the flow of product during various stages of process.

(3) Two handed process chart - (left hand and right hand)

The two handed process chart is a process chart in which the activities of worker's hand (or limbs) are recorded in their relationships to one another. This chart is used to improve the motion sequence of an operator. This chart is used to minimise the necessary motion and to arrange the remaining motion in economic way. The symbols are:

○ - Operation

⇒ Transport

D - Delay

▽ - Hold

□ - Inspection

⊕ - Operation cum transport

(4) Outline process charts - It gives the idea of various operation, inspection and storage done in sequence for all the component that go into a particular product or assembly.

Inventory Control

It is the process of keeping the right number of parts and products in stock to avoid shortage, overstocks and other costly problems. Inventory control focuses on cutting the number of slow selling product, This saves business time and money.

Inventory control also known as stock control

Types of Inventory

- (1) Raw material
- (2) Work in progress
- (3) MRO inventory
- (4) Finished goods
- (5) Vendor Managed inventory
- (6) pipeline & decoupling inventory

Economic Order Quantity (EOQ)

It is the optimum inventory should purchased to minimize the costs of ordering and holding

$$EOQ = \sqrt{\frac{2AP}{C}}$$

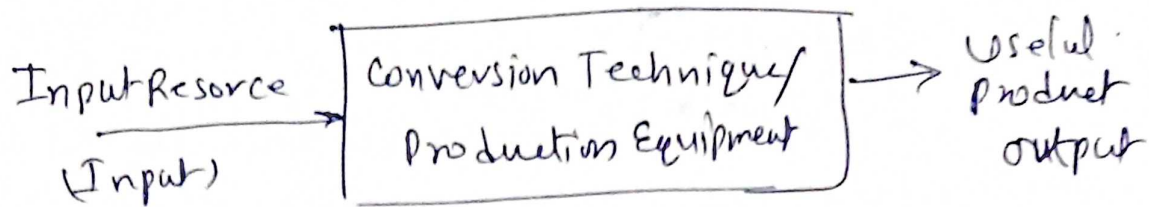
A = Total item consumed

P = Procurement cost per order

C = Annual Inventory carrying cost

Reorder point formula

It determine the right time to order more stock, calculating this means adding together your lead time demand in days and safety stock in a day.



Activities of Industrial Engineering.

- To aid -
- (1) selection of process & assembly method
 - (2) selection of tool equipment & design
 - (3) facility plant location layout material handling etc
 - (4) system of production inventory, quality distribution
 - (5) develop & install job evaluation system
 - (6) develop quality specifications
 - (7) Incentive to worker for motivate
 - (8) Operation research.
 - (9) man machine system
 - (10) cost control system analysis
 - (11) Training programme.

Techniques of Industrial Engineering

(1) Site selection and plant Design Technique:-
depend upon availability, cost, raw material, consumer transportation water electricity labour climatic condition etc

(2) Method Engineering Techniques

- six features 'what, why, How, whose, when, who'.
- analysis the normal time & standard time.

- (1) Method study
- (2) Time study
- (3) motion economy

(3) Organization & planning Technique

all business function in systematic way, customer satisfaction, employment, welfare activity, production planning, control Inventory control.

(4) Commercial Engg. Techniques

- Job evaluation - wage plan & incentive
- Recruitment training purchasing Value Engg.
- Economic analysis.

(5) operation Research Technique:-

- Linear programming, simulation, queuing, Network analysis
- Assignment, transportation Dynamic programming

(6) Statistical Techniques

To product reliability for evaluate $\bar{X}$, c-chart
CPM, PERT

Manufacturing Technology

Manufacturing technology is a term that can refer to a number of modern method of science, production and Engineering that assist in industrial production and various manufacturing processes."

It is detailed study of various processes used to produce different parts and to assemble them into a machine

The word "manufacturing" is derived from latin word "manus" which means hand and fact means made i.e. made by hand.

It is the steps through which raw material are transformed into final product

Manufacturing process based on Technology classified as:-

(1) Casting process.

Sand casting die casting, centrifugal casting Investment casting Shell casting

(2) De forming process

High Energy rate forming (HERF) High velocity forming HVF etc

(3) Metal machining process

Turning, milling, shaping, drilling, bending, drawing etc

(4) Joining Processes

welding, brazing, soldering, riveting, bolting, bonding etc

(5) Heat treatment process

Annealing, normalizing hardening tempering etc.

(6) Plastic Manufacturing process

compression moulding, injection moulding transfer moulding

(7) New Metal Machining processes

ECM, EDM, AJM, LBM etc

Selection of manufacturing process

It depends on the following:-

- (1) Type of raw material
- (2) Nature of material
- (3) Volume of material
- (4) Quality and properties of material
- (5) Complexity of product
- (6) Economic factor.

Role of computers in manufacturing

- (1) NC, CNC & DNC
- (2) Computer Aided Design (CAD)

- (3) Computer Aided Manufacturing (CAM)
- (3) Computer Aided process planning (CAPP)
- (5) Product, Planning and control (PPC)
- (6) Flexible Manufacturing system (FMS)
- (7) Micro ElectroMechanical system
- (8) Industrial Robotics
- (9) Material Requirement planning
- (10) Auto guided Vehicle.

Artificial Intelligence (AI) and machine learning seem to be the next wave in manufacturing. AI is helping production team, analyse data and use the insight to replace inventory, reduce operational costs and offer seamless quality control over the entire manufacturing process.

Lathe

A lathe is a machine tool that rotates a workpiece about an axis of rotation to perform various operations. This machine is also known as "mother of all machines".

A lathe machine is a machine tool that removes the undesired material from a rotating workpiece in the form of chips with the help of a tool that is traversed across the work and can be fed deep into the work.

Lathe machine parts :-

- (1) Bed
- (2) Headstock
- (3) Tailstock
- (4) Carriage
- (5) Saddle
- (6) Crossslide
- (7) Compound Rest
- (8) Tool post
- (9) Apron
- (10) Chuck
- (11) Head screw
- (12) Spindle

Lathe Working Principle

A lathe works on the principle of rotating the workpiece and a fixed cutting tool.

Types of Lathe

- (1) Engine Lathe
- (2) Turret lathe
- (3) Capstan lathe
- (4) Toolroom lathe
- (5) Bench lathe
- (6) CNC lathe
- (7) Speed lathe.
- (8) Automatic lathe machine

operation performed on lathe

- (1) Turning
- (2) Facing
- (3) Threading
- (4) Parting

- (5) Knurling
- (6) Drilling
- (7) Boring
- (8) Reaming
- (9) Chamfering

Drilling Machine

It is another type of production machine in which works are to drill the workpiece. Drilling is single purpose machine tool used to make hole in workpiece. The hole are being made on the workpiece by making ~~the~~ use of rotating tool called drill bit or twist drill.

Working Principle of Drilling.

That the rotating edge of the tool exerts a large force on the workpiece and holes are created in the workpiece.

Parts :-

- (1) Bed
- (2) Pillar
- (3) Swivel table
- (4) Stepped pulley
- (5) Spindle
- (6) Drill bit
- (7) Handwheel
- (8) Motor

Drilling machine specifications

- (1) maximum diameter of drill
- (2) length of arm and column diameter.

$$\text{Cutting speed} = \frac{\pi \times D \times N}{1000} \text{ meter/minute}$$

$$\text{Machining time} = T = \frac{L}{N \times f} \text{ minute.}$$

$f = \text{feed}$
 $N = \text{rpm of drill}$
 $L = \text{length of the hole.}$

Drilling machine operations

- (1) Drilling
- (2) Boring
- (3) Reaming
- (4) Tapping
- (5) Honing
- (6) Spot Facing
- (7) Counter Boring and Counter sinking.

Drilling machine types

- (1) Portable
- (2) Sensitive
- (3) upright or column
- (4) Gang drilling
- (5) Automatic drilling machine
- (6) Turret drilling
- (7) Radial drilling machine.

Advantage of drilling machine

- (1) High speed
- (2) High output
- (3) Easy to operate
- (4) High flexibility
- (5) low maintenance and lower cost



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DETAILED LECTURE NOTES

Campus: Course:

Class/Section:

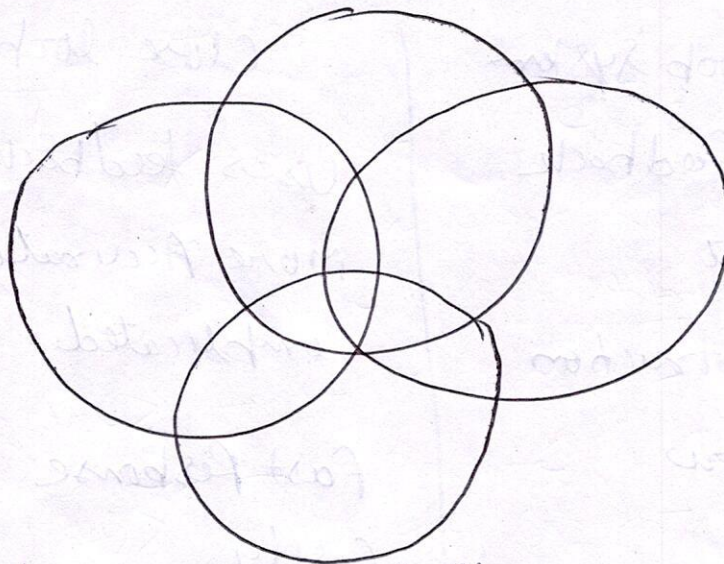
Date:

Name of Faculty:

Name of Subject:

Code:

Mechatronics: is the synergistic integration of mechanics and Mechanical Engineering, Electronics, Computer Technology and IT to produce or enhance products & System.



Actuators/Sensors



Signals and Conditioning



Digital Logic System



Software and Data acquisition System



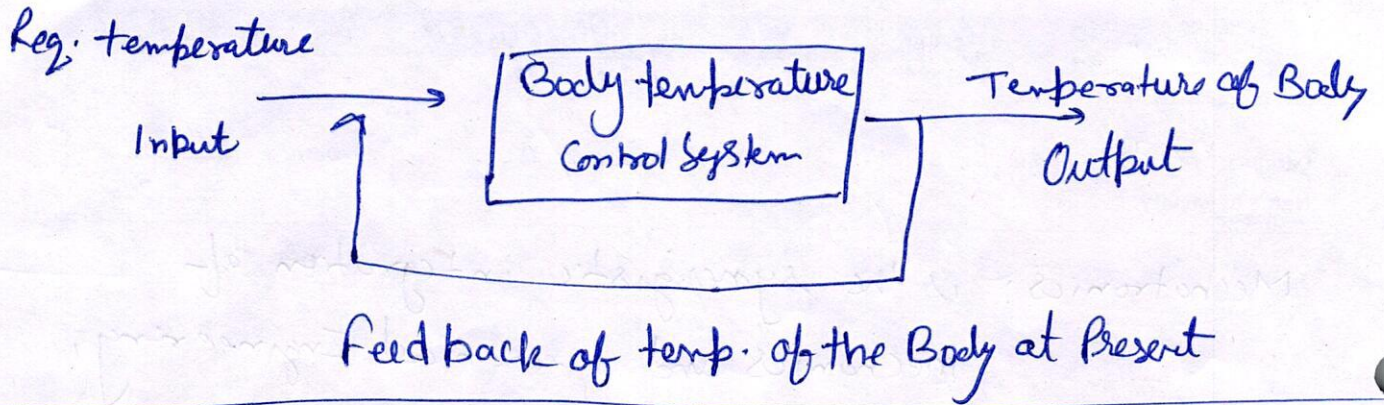
Computer & Display Services

Examples of Mechatronics System

- NC, CNC machine tools
- Robotics
- Laser Printers
- Automatic washing machine
- ATM
- Digital watch

Robotics + Automation

- A Control System refers to a group of physical components connected or related in such a manner as to command direct or regulate itself or another system.



Types

- Open Control System (output is not used as a variable)
- Feedback / Closed loop control system

Open loop system

- Does not use feedback
- less Accurate
- Simple in Construction
- Response is slow
- Cheap
- Easy to maintain

Closed loop system

- uses feedback system
- More Accurate.
- Complicated.
- Fast Response
- Costly
- Maintenance Cost high.

In Robotics-

- Provide stability, precision, and Automation.

Examples: Trajectory Control, Speed regulation, Position Control



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Motors in Robotics:-

Role - Provide rotational or linear motion required for robot movement

Types of motors

DC Motors - Simple, high torque at low speeds.
- Used in mobile robots.

Stepper Motors - Moves in discrete steps
- Excellent for precise position control

Servo Motors -

DC Motor with position feedback
High precision in angular motion

AC Motors - Robust, efficient for industrial robots

Actuators:-

Device which convert Energy into Mechanical motion

Types of Actuators:-

- Electrical Actuators - DC, stepper, servo motors
- Pneumatic Actuators - use compressed air, fast response, lightweight
- Hydraulic Actuators - use pressurized liquid, high force output
- Smart Actuators - Integrate sensors and controllers for Adaptive Control.

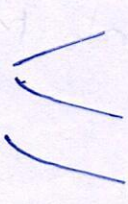
Sensors in Robotics - Provide feedback to control systems; enable robots to perceive the environment.

Types

- Position Sensors:- Encoders, Potentiometers.
- Velocity Sensors - Tachometers.
- Force & Torque Sensors - Measure gripping force, load handling.
- Proximity Sensors - Infrared ultrasonic, inductive, capacitive
- Vision Sensors - Cameras for image recognition, object tracking
- Tactile Sensors - Detect texture, pressure, and touch.

End effectors:- The tool/device at the end of a robot manipulator that interacts with the environment.

Types

Grippers  Mechanical Grippers (Jaws, fingers)
Vacuum Grippers (Suction)
Magnetic Grippers (ferrous materials)

- Control System ensure accuracy and stability.
- Motors & actuators provide the motion/force for robotic tasks.
- Sensors enable awareness and adaptability
- End effectors determine how robots interact with the physical world.



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DETAILED LECTURE NOTES

ROBOT

Industrial robots are automated, reprogrammable, multi-purpose manipulators programmed to perform a variety of tasks in an industrial environment, such as welding, assembly, and logistics. They are designed for precision, speed, and repeatability in high-volume production and are employed in industries like automotive, electronics, and food processing, contributing to increased productivity and efficiency.

Key Characteristics

- **Automated and Programmable:** Industrial robots operate automatically and can be reprogrammed to perform different tasks, making them versatile.
- **Multi-purpose Manipulator:** They have an arm or manipulator that can perform various actions.
- **Multiple Axes:** Robots move on three or more axes, allowing for complex and precise movements.
- **Fixed or Mobile:** They can be fixed in place or mounted on a mobile platform.

Robot configuration refers to the specific arrangement and positions of a robot's joints, which defines the unique pose (position and orientation) of its end effector in space. There are several standard robot configurations, including Cartesian, Cylindrical, Polar, Articulated (jointed-arm), and SCARA, each with distinct joint types (prismatic for linear movement, revolute for rotation) and workspace shapes, making them suitable for different applications.

Here are the common types of robot configurations:

- **Cartesian Robot:**

Consists of three prismatic joints (linear) that move along the X, Y, and Z axes, providing a rectangular work volume.

- **Cylindrical Robot:**

Features two prismatic joints and one revolute (rotary) joint, creating a cylindrical workspace.

- **Polar Robot:**

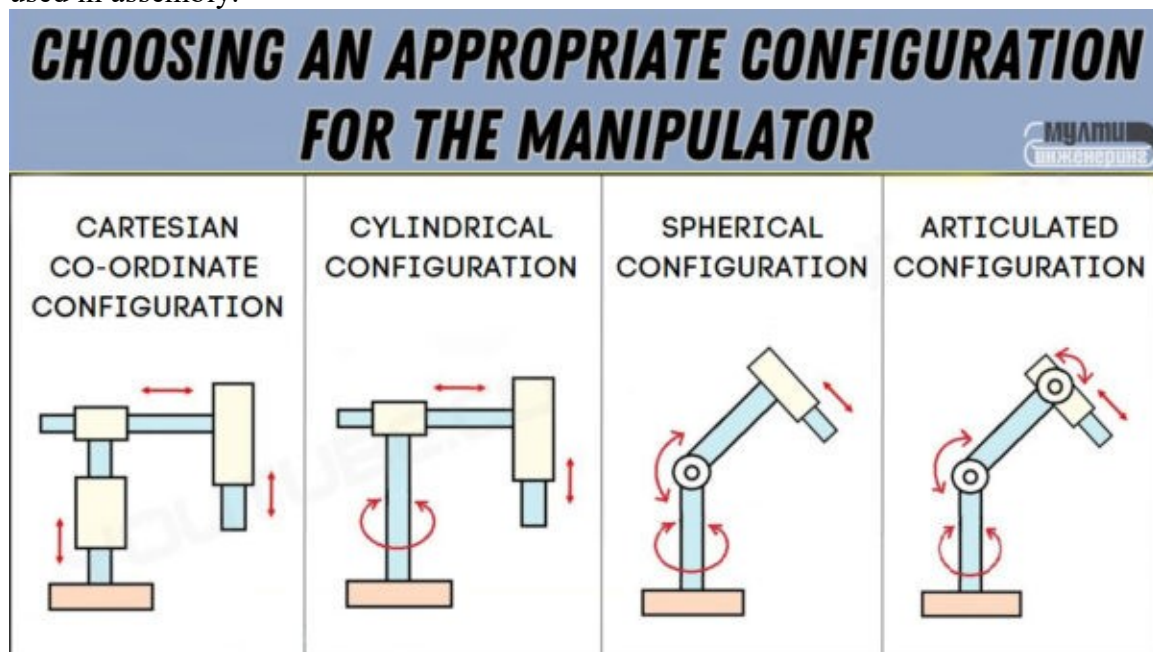
Uses a combination of prismatic and revolute joints to create a spherical work volume, with a rotating base, a rising/lowering prismatic joint, and a revolute arm.

- **Articulated Robot (or Jointed-Arm Robot):**

Resembles a human arm with several revolute joints, offering high flexibility and a large, complex work envelope.

- **SCARA Robot:**

Selective Compliance Assembly Robot Arm, with two parallel revolute joints that allow for movement in a horizontal plane and a prismatic joint for vertical movement, often used in assembly.



Robot Drive Systems

Types of Actuators

*Electric Motors, like: Servomotors, Stepper motors or Direct-drive electric motors

*Hydraulic actuators

*Pneumatic actuators

Electric Drive System:

The electric drive systems are capable of moving robots with *high power* or speed. The actuation of this type of robot can be done by either DC servo motors or DC stepping motors. It can be well –suited for rotational joints and as well as linear joints. The electric drive system will be perfect for *small robots* and precise applications. Most importantly, it has got greater accuracy and repeatability. The one disadvantage of this system is that it is slightly costlier. An example for this type of drive system is *Maker 110 robot*.

Hydraulic Drive System:

The hydraulic drive systems are completely meant for the *large –sized robots*. It can deliver high power or speed than the electric drive systems. This drive system can be used for both linear and rotational joints. The rotary motions are provided by the rotary vane actuators, while the linear motions are produced by hydraulic pistons. The *leakage* of hydraulic oils is considered as the major disadvantage of this drive. An example for the hydraulic drive system is *Unimate 2000 series robot*.

Pneumatic Drive System:

The pneumatic drive systems are especially used for the *small type robots*, which have less than five degrees of freedom. It has the ability to offer fine accuracy and speed. This drive system can produce rotary movements by actuating the rotary actuators. The translational movements of sliding joints can also be provided by operating the piston. The price of this system is *less* when compared to the hydraulic drive. The drawback of this system is that it will not be a perfect selection for the *faster operations*.

Robotic control systems are the "brains" of robots, combining hardware and software to perceive their environment, make decisions, and execute commands.

Key Components

- **Sensors:**
The "eyes" and "ears" of a robot, they gather data about the environment and the robot's own state, including position, orientation, forces, and pressure.
- **Actuators:**
These are the components that enable movement and action, such as motors or hydraulic systems that drive the robot's joints and end-effectors.
- **Controllers:**
The central processing unit that receives input from sensors and user commands, processes this data using specific algorithms, and sends commands to the actuators.
- **Software Algorithms:**
The intelligence behind the control system, these algorithms process sensor data, make real-time decisions, and generate commands to achieve desired tasks.

Types of Control Systems

- **Open-Loop Control:**
Relies on pre-programmed instructions without using feedback to correct errors.

- **Closed-Loop Control:**
Uses feedback from sensors to monitor the robot's performance and make real-time adjustments to ensure it achieves its goal.
- **Reactive Control:**
Robots respond directly to immediate stimuli in their environment.
- **Deliberative Control:**
Involves planning and reasoning to determine the robot's actions, often used for more complex tasks.

ROBOT END EFFECTORS

End effectors are devices attached to the end of a robot's arm to help it interact with the surrounding environment. End effectors are essential to robotic systems as they handle, manipulate, and sense objects. These devices are the primary means by which robots interact with their environment and perform useful work. End effectors can be customized to suit specific applications and can be designed to perform a wide range of tasks, including gripping, cutting, welding, painting, and more.

End effectors can be broadly classified into three categories based on their design, construction, and application.

Grippers

Grippers are used to grasp and hold objects securely. They are one of the most commonly used types of end effectors in robotics and are an essential component of many automated systems. The primary function of grippers is to provide robots with the ability to pick up, move, and manipulate objects in a controlled and repeatable manner.

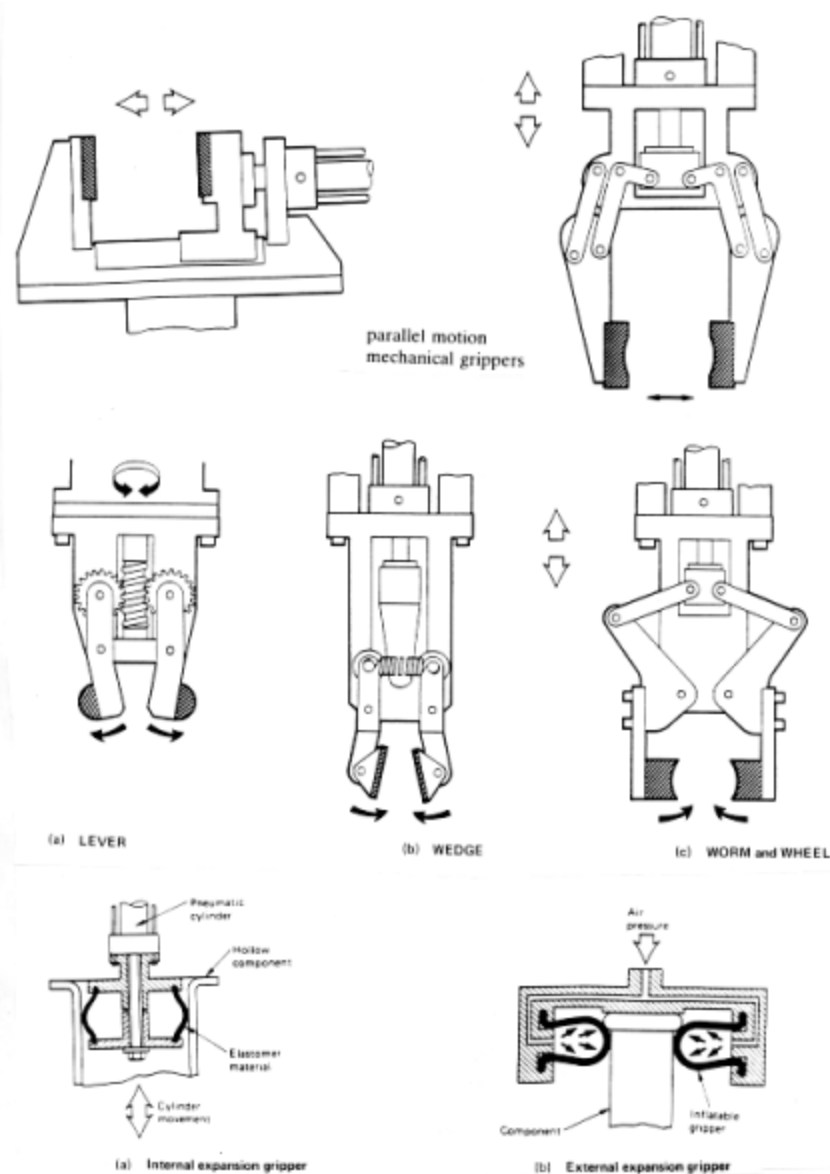
Mechanical Grippers: Mechanical grippers are the most basic type of grippers and work by using jaws or fingers to grab an object. They are simple, robust, and reliable and are often used in applications that require a firm and stable grasp, such as pick-and-place operations or assembly tasks. Mechanical grippers can be further classified based on the type of jaws or fingers used, such as parallel grippers, angular grippers, or three-finger grippers.

Vacuum Grippers: Vacuum grippers use suction cups to grab and hold things. They are primarily used in applications that require handling flat, smooth, or porous objects, such as glass panels or printed circuit boards. Vacuum cups are easy to use, have a fast cycle time, and are highly reliable. However, they may not be suitable for handling irregularly shaped or heavy payloads.

Magnetic Grippers: Magnetic grippers use magnetic fields to hold ferromagnetic items such as steel plates. They are used in applications that require handling heavy or irregularly shaped objects that cannot be easily managed with mechanical or vacuum

grippers. Magnetic grippers are highly reliable, require minimal maintenance, and have a fast cycle time. However, they may not be suitable for handling non-ferromagnetic materials.

Servo Grippers: Servo grippers use motors and gearboxes to control the gripping force and positioning of the robot precisely. They are used in applications that require high precision and flexibility, such as quality control or inspection tasks. Servo grippers can be programmed to adjust their gripping force and position based on the size and shape of the object being handled, making them highly versatile.



Sensors

Sensors are essential components of robotic systems, providing robots with the ability to gain a perception of their environment. Sensors gather data about the robot's surroundings, including the position and orientation of objects around it and the robot itself. The robot's control system then uses this information to decide how to interact with its environment.

Proximity sensors: Proximity sensors are used to detect the presence or absence of objects in the proximity of the end effector. These sensors can detect various targets, including metal, plastic, and even liquids.

Force/torque sensors: Force/torque sensors measure the amount of force or torque being applied to the robot or to the objects it interacts with. These sensors are often used in robotic grippers to measure the force required to grip an object or in robotic arms to measure the force required to move an object.

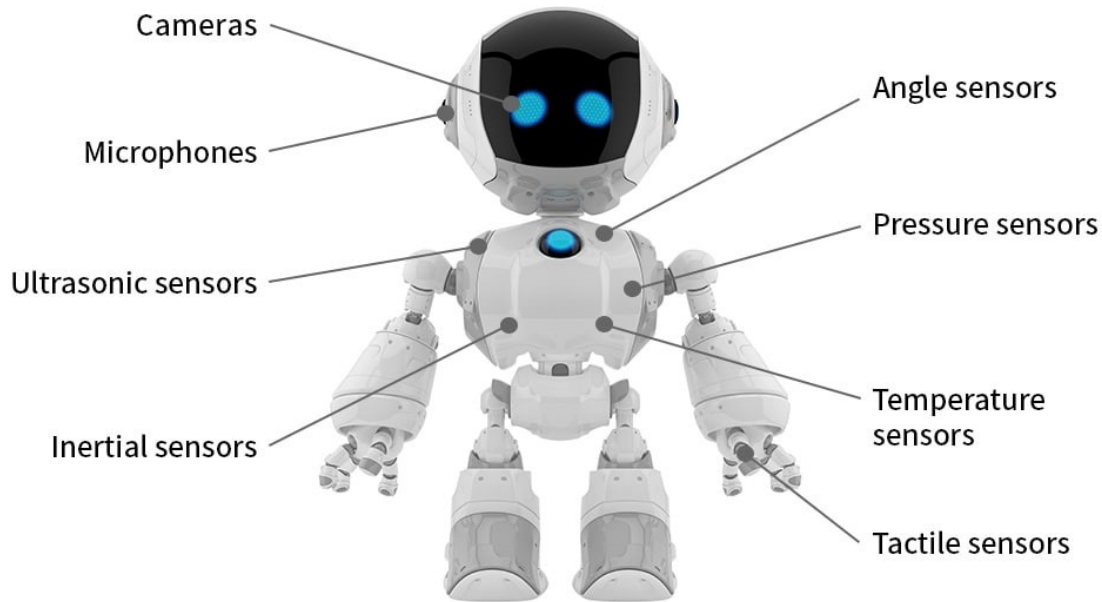
Cameras: Vision sensors use cameras and other imaging technologies to provide visual feedback on the position, orientation, and movement of the end effector and the objects it interacts with. These sensors are often used in applications requiring precise positioning and manipulation. Cameras are also used for object recognition, tracking, navigation, and obstacle avoidance. Cameras for end effectors can even have 3D and thermal imaging capabilities if the application demands.

Light sensors: Light sensors can detect the presence or absence of light and are used to provide feedback on the position of objects in low-light environments.

Magnetic sensors: Magnetic sensors or hall effect sensors are used to detect the presence of magnetic fields. They can detect the position and orientation of objects that contain magnets.

Range Sensors: Range sensors are used to measure the distance between the robot's end effector and objects near it.

The advantages of sensors include increased accuracy, efficiency, and flexibility.



Applications and Industries

- **Automotive:** A primary application, used for assembly, welding, and other manufacturing processes.
- **Logistics:** Sorting and transporting goods in warehouses.
- **Electronics:** Mounting components on circuit boards and other high-speed tasks.
- **Food & Beverage:** Filling, packaging, and handling food products.
- **Biomedical:** Performing high-precision motions in research and pharmaceutical settings.

Benefits

- **Increased Productivity:** Robots can operate 24/7 without breaks, leading to continuous production.
- **Improved Precision and Repeatability:** They can perform tasks with consistent accuracy.
- **Enhanced Safety:** Robots can take over dangerous or repetitive tasks that are risky for humans.
- **Cost Savings:** Automation can lead to reduced labor costs and higher productivity, improving a company's bottom line.

Engineering Materials

The selection of a specific material for a particular use is a very complex process. However, one can simplify the choice if the details about (i) Use parameters, (ii) manufacturing processes, (iii) functional requirements, and (iv) cost considerations are known. While selecting materials for engineering purposes, properties such as impact strength, tensile strength, and hardness indicate the suitability for selection.

Mechanical properties of Engineering Materials

Tensile Strength: This enables the material to resist the application of a tensile force. The internal structure of the material provides the internal resistance to withstand the tensile force. Ultimate strength is the unit stress; measures in kgf per square millimeter, developed in the material by the maximum slowly applied a load that material can withstand without rupturing in a tensile test.

Shear Strength:

It is the ability of a material to resist the shear force applied to the material.

Compressive Strength:

It is the ability of a material to withstand pressures acting on a given plane.

Elasticity:

It is the property of material due to which it returns to its original shape and size after releasing the load. Any material that is subjected to an external load is distorted or strained. Elastically stressed materials return to their original dimensions when the load is released.

Hardness:

It is the degree of resistance to indentation, scratching, abrasion, and wear. Alloying techniques and heat treatment help to achieve the same.

Ductility:

This is the property of a metal by virtue of which it can be drawn into wires or elongated before rupture takes place. It depends upon the grain size of the metal crystals.

Malleability:

It is the property of a metal to be deformed or compressed permanently into the sheet without fracture. It shows the ability of the material to be rolled or hammered into thin sheets.

Impact Strength:

It is the energy required per unit cross-sectional area to fracture a specimen, i.e., it is a measure of the response of a material to shock loading.

Toughness:

It is the ability of the material to absorb energy before fracture or rupture. It may be presented as impact strength of the material.

Brittleness:

The term “brittleness” implies sudden failure. It is the property of breaking without warning, i.e., without visible permanent deformation.

Wear Resistance:

The ability of a material to resist friction wear under particular conditions, i.e., to maintain its physical dimensions when in sliding or rolling contact with a second member.

Corrosion Resistance:

Those metals and alloys which can withstand the corrosive action of a medium, i.e., corrosion processes proceed in them at a relatively low rate that are termed as corrosion-resistant.

Density:

This is an important factor of a material where weight and thus the mass is critical, i.e., aircraft components.

Classification of Engineering Materials

- (i) Metals (ferrous and non-ferrous) and alloys
- (ii) Ceramics
- (iii) Organic polymers
- (iv) Composites
- (v) Semi-conductors
- (vi) Biomaterials
- (vii) Advanced materials

Ferrous Metals

In ferrous materials, the main alloying element is carbon (C). Depending on the amount of carbon present, these alloys will have different properties, especially when the carbon content is either less/higher than 1.5%

Low Carbon Steel:

It contains up to 0.3% carbon and 1% manganese. Since its microstructure consists of ferrite and pearlite so it is relatively softer than the other carbon steel. It has good ductility and toughness. Mild steel (Carbon % 0.15–0.3) due to its good strength, high machinability, and weldability property is an extensively used engineering material. It is used as different structural sections (channel, angles, etc.), sheets, automobile components, etc.

Medium Carbon Steels:

These contain carbon between 0.3% and 0.6%. The strength of these materials is high but their weldability is comparatively less. Due to higher C%, it can be heat treated to get higher hardness. It is used as railway track and wheels, crankshafts, gears, etc.

High Carbon Steels:

These contain carbon varying from 0.65% to 1.5%. These materials get hard and tough by heat treatment and their weldability is poor. The steel formed in which carbon content is up to 1.5%, silica up to 0.5%, and manganese up to 1.5% along with traces of other elements is called plain carbon steel.

Cast Irons:

The carbon content in these substances vary between 2% and 4%. The cost of production of these substances is quite low and these are used as ferrous casting alloys.

Gray Cast Iron:

These alloys consist of carbon in form of graphite flakes, which are surrounded by either ferrite or pearlite. Because of the presence of graphite, fractured surface of these alloys look grayish and so is the name for them. The alloying addition of Si (1–3 wt%) is responsible for decomposition of cementite, and also high fluidity. Thus castings of intricate shapes can be easily made. Due to graphite flakes, gray cast irons are weak and brittle. However, they possess good damping properties, and thus typical applications include: base structures, bed for heavy machines, etc. they also show high resistance to wear.

White Cast Iron:

When Si content is low (<1%) in combination with faster cooling rates, there is no time left for cementite to get decomposed, thus most of the brittle cementite retains. Because of the presence of cementite, the fractured surface appears white, hence the name. They are very brittle and extremely difficult to machine. Hence their use is limited to wear resistant applications such as rollers in rolling mills. Usually, white cast iron is heat treated to produce malleable iron.

Nodular (or Ductile) Cast Iron:

Alloying additions are of prime importance in producing these materials. Small additions of Mg/Ce to the gray cast iron melt before casting can result in graphite to form nodules or sphere-like particles. Matrix surrounding these particles can be either ferrite or pearlite depending on the heat treatment. These are stronger and ductile than gray cast irons. Typical applications include pump bodies, crankshafts, automotive components, etc.

Malleable Cast Iron:

These formed after heat treating white cast iron. Heat treatments involve heating the material up to 800–900°C, and keep it for long hours, before cooling it to room temperature. High temperature incubation causes cementite to decompose and form ferrite and graphite. Thus these materials are stronger with appreciable amount of ductility. Typical applications include: railroad, connecting rods, marine, and other heavy duty service

Alloy Steel

The common alloying elements are Chromium, Nickel, Molybdenum, Tungsten, Cobalt, Copper, Manganese, Silicon, and Sulfur, Phosphorous, etc. Depending on the percentage of alloying elements, mechanical properties like strength, hardness corrosion, etc. changes under different operating conditions.

Effect of Alloying Elements**Manganese (Mn) –**

When Manganese percentage exceeds a normal percentage (1.65%) in steel, then the steel is known as Manganese steel. Manganese up to 1.95% improves hardness, tensile strength, and hot working property.

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Silicon (Si) –

The steel having more than 0.6% Si is known as Silicon steel. It acts as deoxidizer and graphitizer. As it dissolves in ferrite so it is not carbide former. It has high magnetic permeability but very low hysteresis loss. It is extensively used in electrical industries. Steel with 0.5% C and 3–4% Si is used to make a motor, transformer cores. This is known as transformer steel..

Phosphorus (P) –

Small amount (about 0.05%) is present in normal steel. If it is increased to 0.12% in low carbon steel then it improves strength, hardness, corrosion resistance, machine ability, etc.

Nickel (Ni) –

When added up to 5% it improves static and impact load bearing properties. Higher % of Nickel addition improves corrosion resistance. Steels with 1.5–3% Ni are suitable for loco boilers, railway axles, etc.

Chromium (Cr) –

Chromium addition in plain carbon steel improves hardenability, strength, wear resistance, corrosion, and red hot resistance if added more than 4% it improves corrosion resistance. Chromium improves hardenability It is widely used in tool steel.

Tungsten (W) –

The steel retains its hardness at a higher temperature with addition of tungsten as an alloy. It is a strong carbide former. Tungsten carbide is extremely hard and stable. It improves wear and abrasive resistance in steel. It retards softening of Martensite during tempering and gives hot hardness. So is commonly added to make tool and hot-working die steel. In High-Speed Steel (HSS), it is added up to 18%.

Molybdenum (Mo) –

The steel becomes more wear resistant with the addition of Molybdenum. Its red hot hardness is high as compared to Carbon steel as Molybdenum Carbide can withstand higher temperature as compared to iron carbide. It is used to make aircraft components, pressure vessels, and springs. 5% Mo is added in some HSS steel.

Vanadium (V) –

It improves red hot hardness, fatigue resistance and wear resistance property. Vanadium carbide has the highest hardness and wear resistance property amongst alloying elements added to steel. 2% Vanadium is added in some HSS.

Non-Ferrous Metals

Aluminum Alloys:

Aluminum alloys have high thermal and electrical conductivity and good corrosion resistant characteristics. As Al has FCC crystal structure, these alloys are ductile even at low temperatures and can be formed easily. However, the great limitation of these alloys is their low melting point (660°C), which restricts their use at elevated temperatures. Chief alloying elements include Cu, Si, Mn, Mg, Zn. Recently, alloys of Al and other low-density metals like Li, Mg, Ti gained much attention as there is much concern about vehicle weight reduction. Al-Li alloys enjoy much more attention especially as they are very useful in aircraft and aerospace industries. Common applications of Al alloys include beverage cans, automotive parts, bus bodies, aircraft structures, etc

Copper Alloys: One special feature of most of these alloys is their corrosion resistant in diverse atmospheres. **Brass, alloys** of Cu and Zn where Zn is substitution addition (e.g., yellow brass, cartridge brass, Muntz metal, gilding metal); **Bronze, alloys** of Cu and other alloying additions like Sn, Al, Si, and Ni. Bronzes are stronger and more corrosion resistant than brasses. Mention has to be made about beryllium coppers who possess a combination of relatively high strength, excellent electrical and corrosion properties, wear resistance, Applications of Cu alloys include costume jewellery, coins, musical instruments, electronics, springs, bushes, surgical and dental instruments, radiators, etc.

Magnesium Alloys:

The most sticking property of Mg is its low density among all structural metals. Mg has HCP structure, thus Mg alloys are difficult to form at room temperatures. Hence Mg alloys are usually fabricated by casting or hot working. Major alloying additions are Al, Zn, Mn, and rare earth. Common applications of Mg alloys include hand-held devices like saws, tools, automotive parts like steering wheels, seat frames, electronics like casing for laptops, camcorders, cell phones, etc.

Titanium Alloys:

Ti and its alloys are of relatively low density, high strength and have a very high melting point. At the same time, they are easy to machine and forge. However, the major limitation is Ti's chemical reactivity at high temperatures, which necessitated special techniques to extract. Thus these alloys are expensive. They also possess excellent corrosion resistance in diverse atmospheres and wear properties. Common applications include space vehicles, airplane structures, surgical implants, and petroleum and chemical industries.

Refractory Metals:

These are metals of very high melting points. For example Nb, Mo, W, and Ta. They also possess high strength and high elastic modulus. Common applications include space vehicles, X-ray tubes, welding electrodes, and where there is a need for corrosion resistance.

Plastics

Common organic materials are plastics and synthetic rubbers, which are termed as organic polymers. Other examples of organic materials are wood, many types of waxes and petroleum derivatives. Organic polymers are prepared by polymerization reactions, in which simple molecules are chemically combined into long chain molecules or three-dimensional structures. Organic polymers are solids composed of long molecular chains. These materials have low specific gravity and good strength. The two important classes of organic polymers are:

Thermoplastics:

On heating, these materials become soft and hardened again upon cooling, e.g., nylon, polyethylene, etc.

Thermosetting Plastics:

These materials cannot be resoftened after polymerization, e.g., urea-formaldehyde, phenol formaldehyde, etc. Due to cross-linking, these materials are hard, tough, non-swelling and brittle. These materials are ideal for molding and casting into components. The excellent resistance to corrosion, ease of fabrication into desired shape and size, fine lustre, light weight, strength, rigidity have established the polymeric materials and these materials are fast replacing many metallic components. PVC (Polyvinyl Chloride) and polycarbonate polymers are widely used for glazing, roofing, and cladding of buildings. Plastics are also used for reducing the weight of mobile objects, e.g., cars, aircraft, and rockets. Polypropylenes and polyethylene are used in pipes and manufacturing of tanks.

Thermoplastic films are widely used as lining to avoid seepage of water in canals and lagoons. To protect the metal structure from corrosion, plastics are used as surface coatings. Plastics are also used as main ingredients of adhesives. The lower hardness of plastic

materials compared with other materials makes them subjective to attack by insects and rodents. Because of the presence of carbon, plastics are combustible. The maximum service temperature is of the order of 100°C. These materials are used as thermal insulators because of lower thermal conductivity. Rubber materials are widely used for tire of automobiles, insulation of metal components, toys and other rubber products.

Abrasive Materials

Abrasives are hard, non-metallic, sharp-edged and irregular shaped materials used to remove a small amount of materials by cutting action. It may be used in bonded form or as free particles. It is employed in grinding, polishing, super finishing, buffing, honing operations. Commonly used abrasives are alumina (Al_2O_3), Silicon carbide (SiC), Cubic boron nitride (CBN), and diamond.

Ceramics

Ceramics are compound of metallic and nonmetallic materials. It has properties of high compressive strength, low thermal expansion, high elasticity, high hardness, high wear resistance, and low electrical and thermal conductivity. Ceramics are used for tiles, pottery, sanitary wares (Porcelain). The raw materials used for ceramics are clay having a fine sheet like structure, Kaolin (silicate of aluminum) used as clay, flint, and feldspar.

Silica

It is available in abundance in nature in the form of quartz. Most of the glasses contain more than 50% of silica. It is also used in electric materials to increase the magnetic permeability of the materials. It may be used in the form of silicates of various materials as clay, asbestos, mica, glasses, etc.

Glasses

It is a super cooled amorphous material. It consists of more than 50% silica and other additives such as oxides of aluminum, sodium, calcium, magnesium, titanium, lithium, lead, and potassium. It has applications in windows, containers, lighting instruments, cookware, etc. The availability of various types of glasses is soda-lime glass, lead alkali glass, borosilicate glass, etc.